



MINOR PROJECT - II

ON

Co Work: CoWorking Spaces Management Application

(MCA – II SEMESTER; BATCH 2025-27)



Project Mentors

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Contents

- Problem Description
- Aim and Objectives
- Methodology and Technology Used for Project Development
- Project Classes
- Design Documents (Use Case, Data Flow Diagram, Entity-Relation Diagram, etc.)
- Operating Structure of project
- Screenshots
- Conclusion
- Future Scope
- Bibliography

Problem Description

- The current market lacks a unified platform that aggregates coworking spaces while fostering open competition for transparent price discovery, leaving a significant gap for both consumers and providers. Our project addresses this by offering a comprehensive, user-friendly solution that streamlines the booking process for professionals while providing space owners with integrated tools to efficiently manage their day-to-day operations.

Aim and Objectives

- **Aim:**

- To develop a robust web application that aggregate CoWorking Spaces Market using ReactJS, Java's SpringBoot, MongoDB.

- **Objectives:**

- To allow users to search, select and discover Co Working Spaces around them
- To see the review/ratings and to review/rate Spaces
- Enable users to add smaller services in the form of Add-ons before or during their allotted time.

Methodology and Technology Used

- **Methodology used for Project Development:**

- **Stakeholder Mapping:** Identify needs for both "Space Seekers" (consumers) and "Space Providers" (businesses).
- **Feature Prioritization:** Define core functionalities like real-time search, competitive bidding/pricing, and booking management.
- **Data Modeling:** Determine the document structure for MongoDB to handle varied amenities and dynamic pricing.

Methodology and Technology Used

- **Methodology used for Project Development:**

- **Microservices Architecture:** Decouple the user-facing marketplace from the provider-focused management ERP.
- **API Strategy:** Design RESTful endpoints in Spring Boot for seamless communication between the React frontend and MongoDB.
- **Security Framework:** Implement JWT-based authentication and Role-Based Access Control (RBAC) to separate consumer and provider data.

Methodology and Technology Used

- Technology used for Project Development:
 - Java (JDK 21) – Core Backend programming language
 - SpringBoot (4.0.6) – Backend
 - MongoDB Atlas – Cloud based DBMS
 - ReactJS – Frontend
 - IDEs used: IntelliJ / VSCodiums

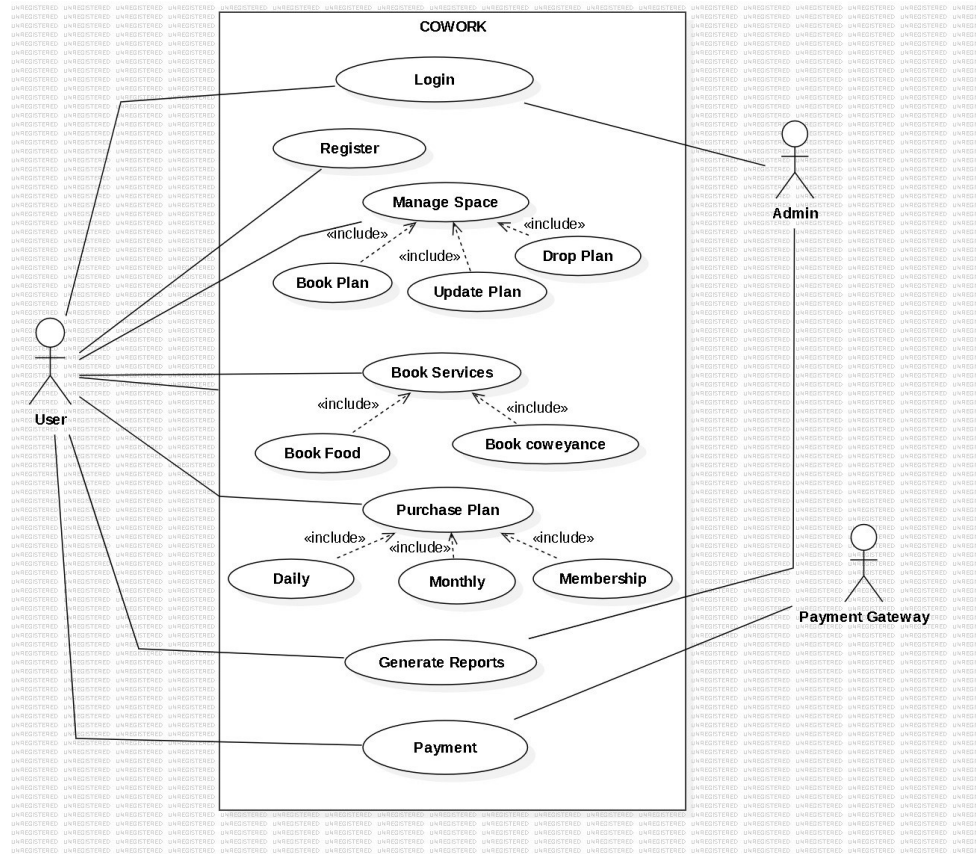
Project Modules

Project Modules:

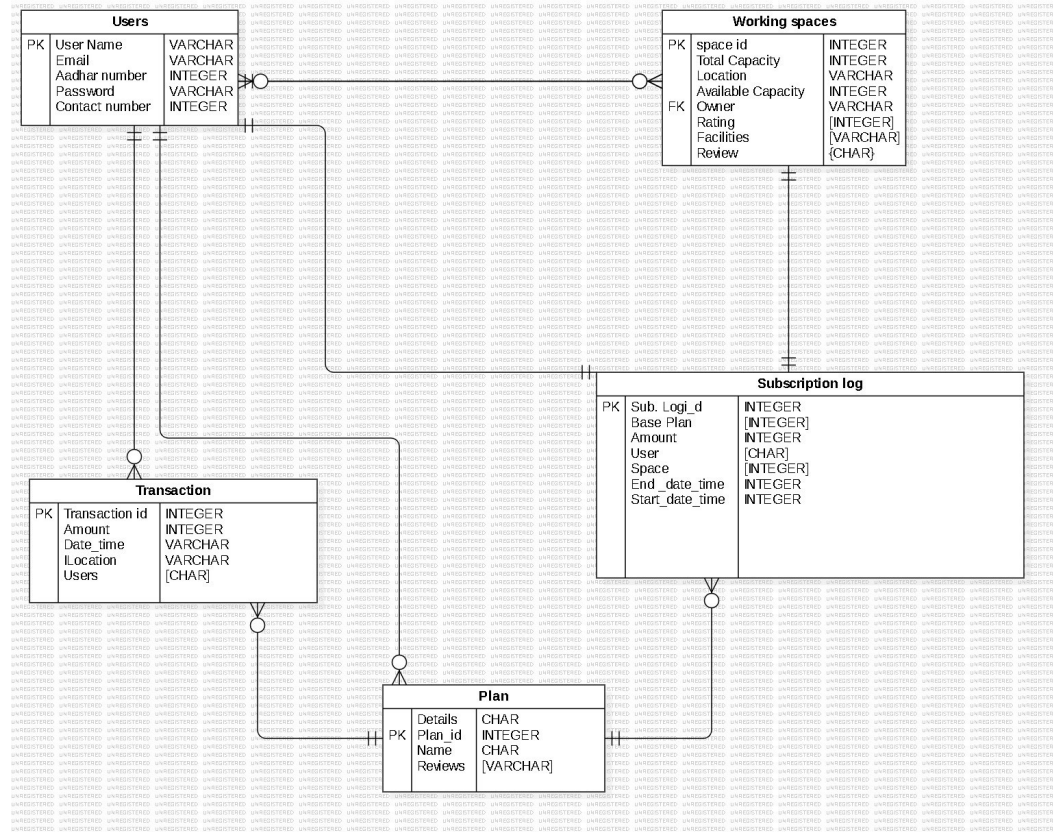
- **User Authentication & Profile Module:** Gateway to the application, managing identities and security.
- **Marketplace & Discovery Module:** "Search Engine" of the platform. (GeoSpatialSearch, Filtering Engine)
- **Booking & Transaction Module:** Handles Real-Time Availability, Pricing Engine and Payment Transactions.
- **Provider Operations (ERP) Module:** The internal tool for space owners to manage their business. (Inventory Management, Operational Dashboard, Maintenance Tracking)

System Design

Use Case Diagram



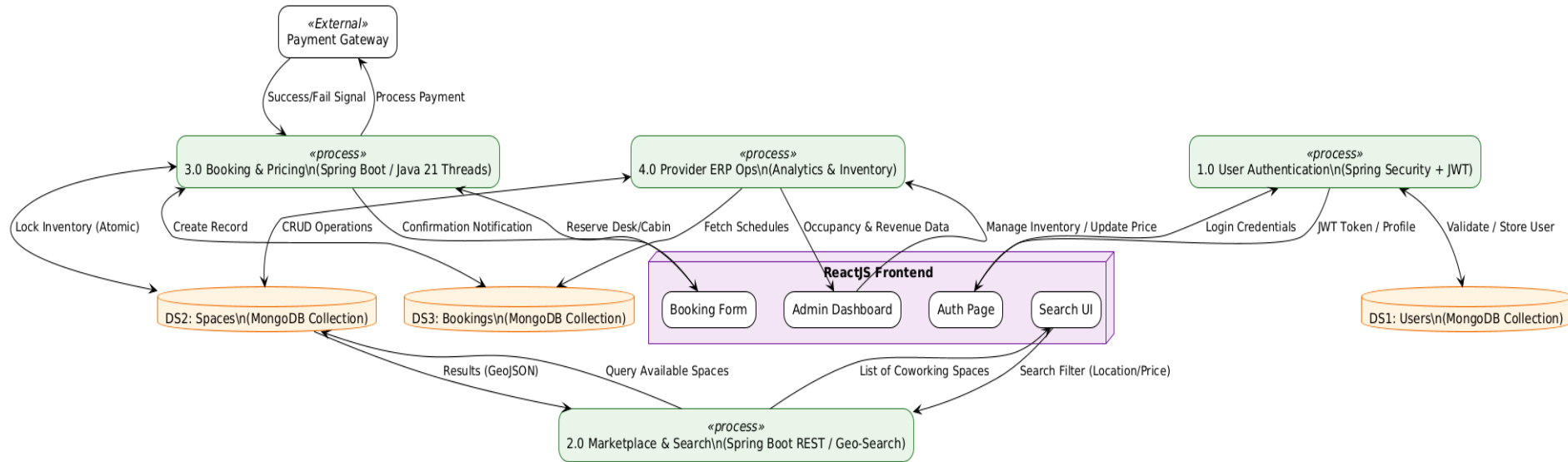
Entity-Relationship (ER) Diagram



Data Flow Diagram (Level - 0)



Data Flow Diagram (Level - 1)

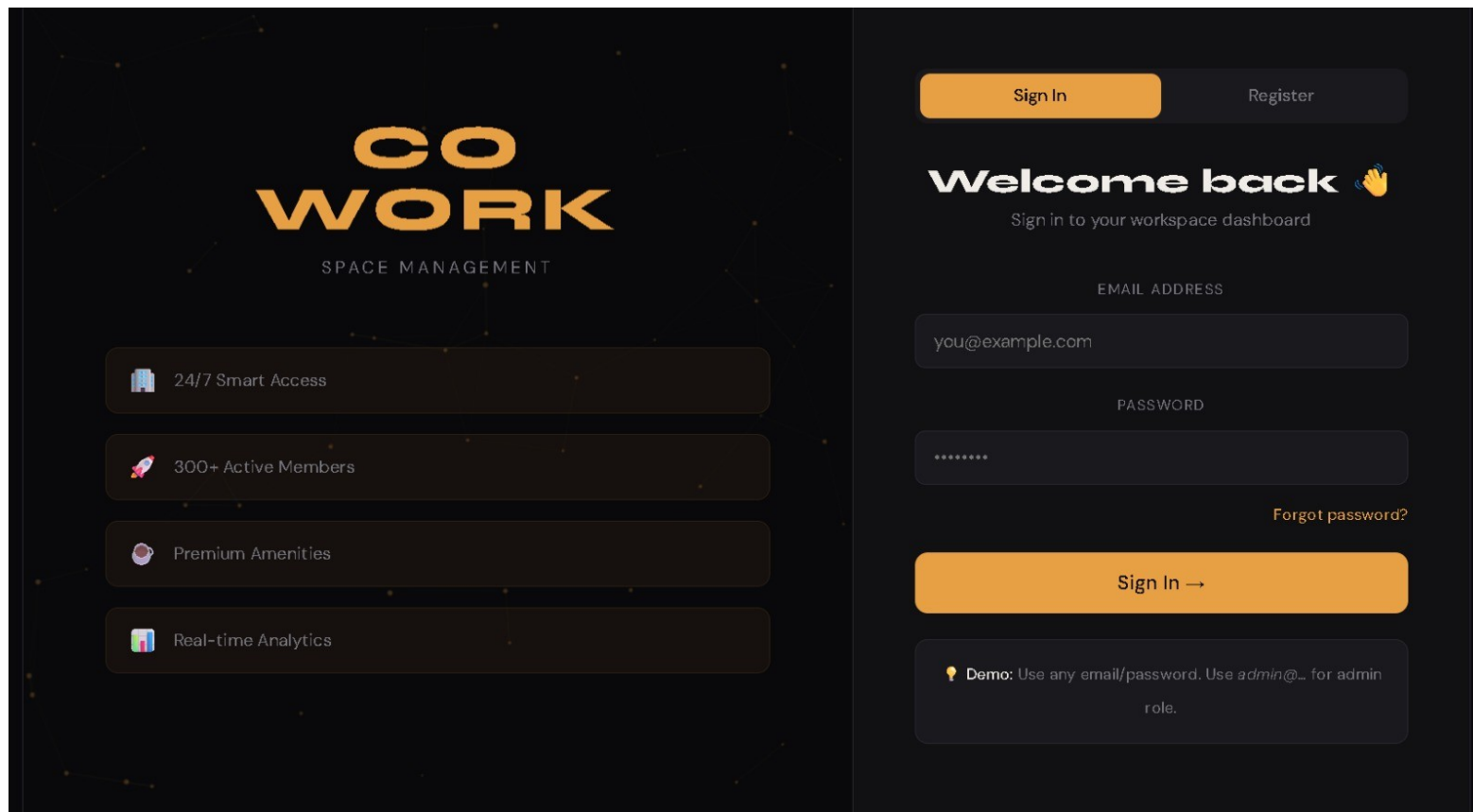


Operating Structure of Project

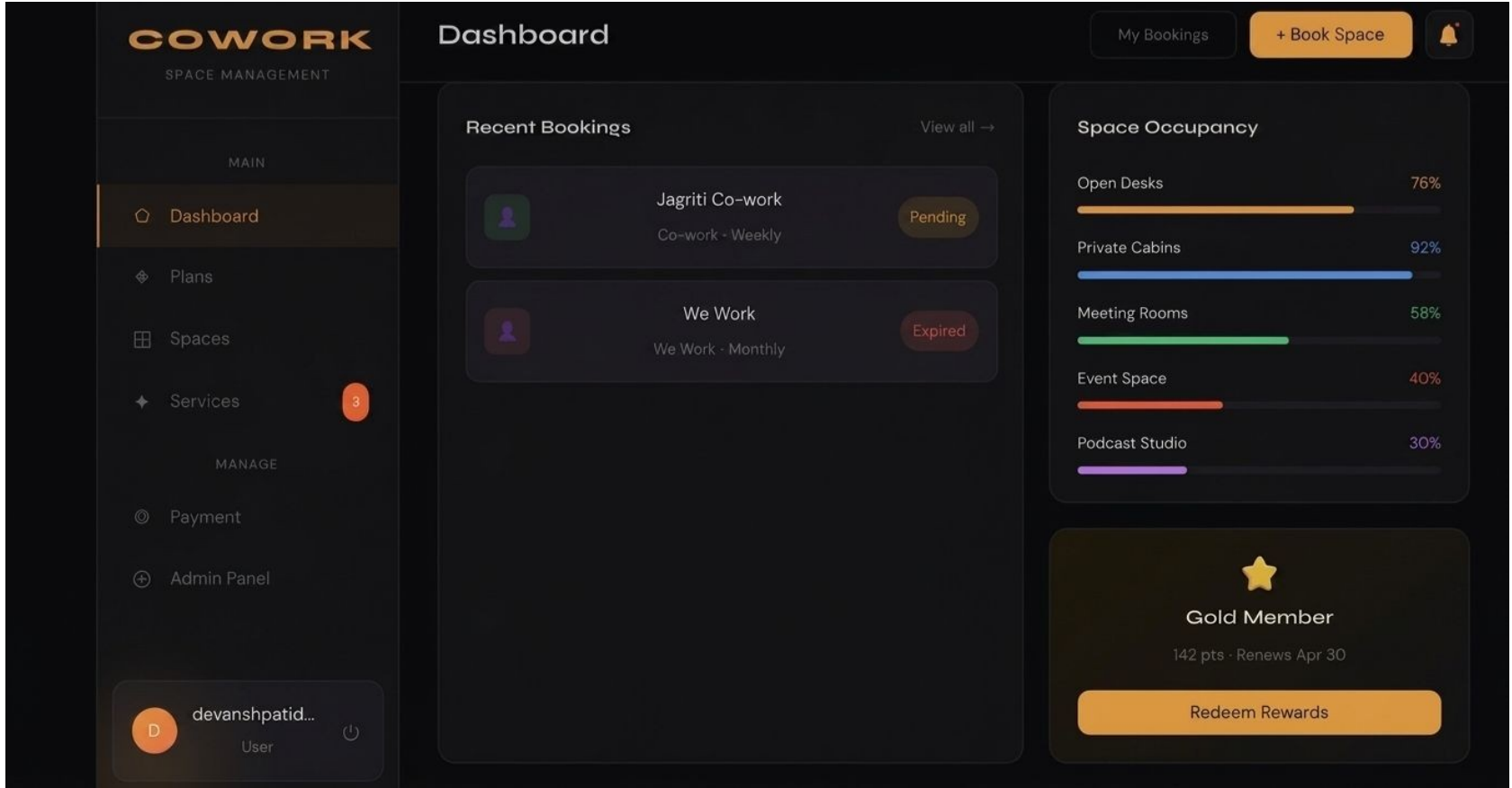
- **API Interaction:** React sends asynchronous HTTP requests via Axios/Fetch to Spring Boot REST endpoints.
- **Data Processing:** Spring Boot handles business logic and interacts with MongoDB, returning data in JSON format.
- **Dynamic Rendering:** React receives JSON responses to update the UI and manage state without refreshing.

Screenshots

Screenshot - 1



Screenshot - 2



The screenshot displays the COWORK SPACE MANAGEMENT Dashboard. The interface is divided into a sidebar, a main content area, and a top navigation bar.

Top Navigation Bar: Includes "My Bookings", "+ Book Space", and a notification bell icon.

Sidebar: Contains the COWORK logo and "SPACE MANAGEMENT". The main menu is categorized into "MAIN" and "MANAGE".

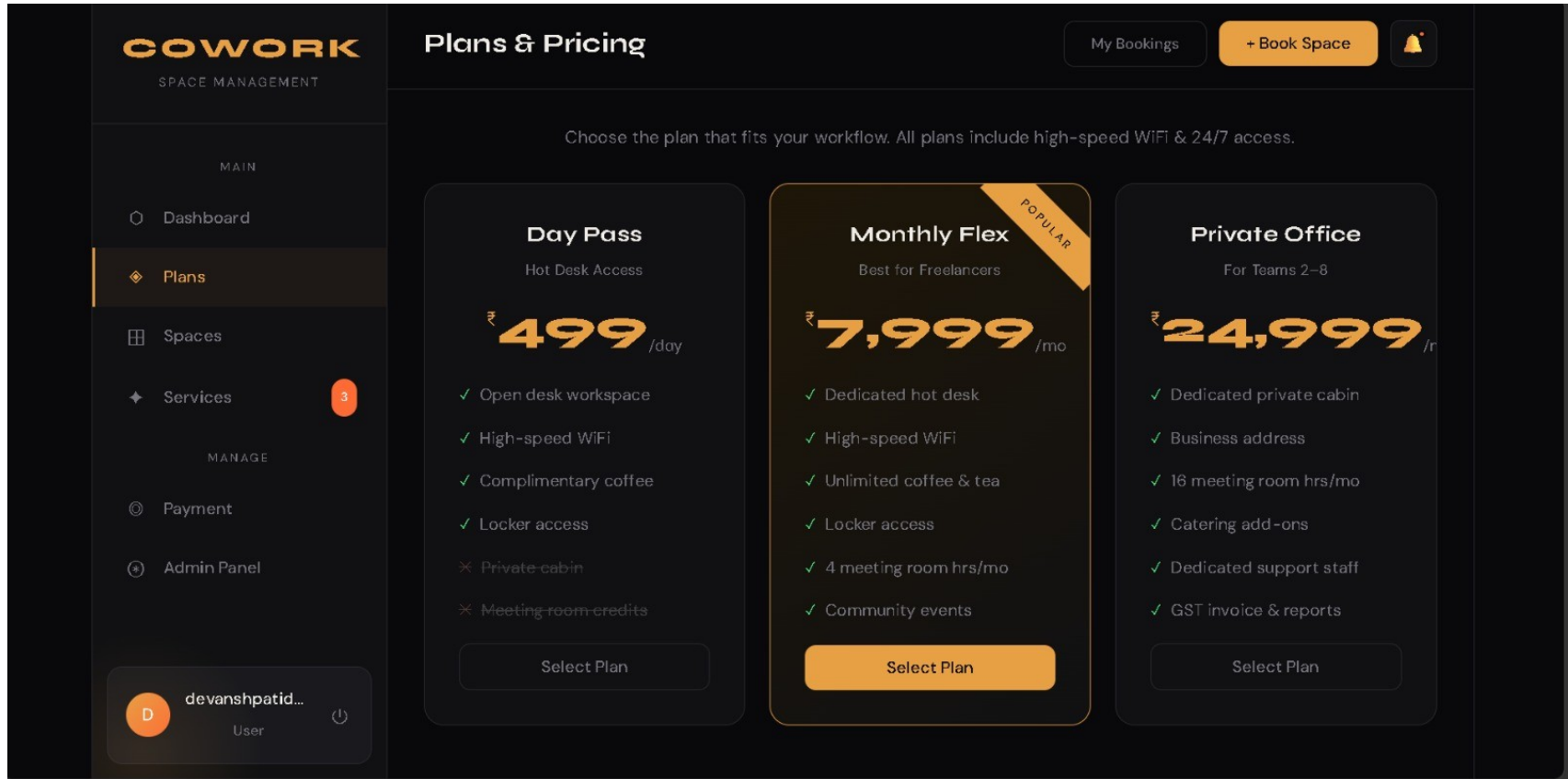
- MAIN:**
 - Dashboard (selected)
 - Plans
 - Spaces
 - Services (3 items)
- MANAGE:**
 - Payment
 - Admin Panel

Main Content Area:

- Recent Bookings:** A list of bookings with a "View all" link.
 - Jagriti Co-work:** Co-work - Weekly, Pending status.
 - We Work:** We Work - Monthly, Expired status.
- Space Occupancy:** A section showing occupancy percentages for various spaces:
 - Open Desks: 76%
 - Private Cabins: 92%
 - Meeting Rooms: 58%
 - Event Space: 40%
 - Podcast Studio: 30%
- Gold Member:** A section highlighting a Gold Member with 142 pts, renewing on Apr 30, and a "Redeem Rewards" button.

User Profile: Located at the bottom left of the sidebar, showing the user "devanshpatid..." with a profile picture and a power button.

Screenshot - 3



COWORK
SPACE MANAGEMENT

Plans & Pricing

Choose the plan that fits your workflow. All plans include high-speed WiFi & 24/7 access.

Day Pass
Hot Desk Access
₹ **499** /day

- ✓ Open desk workspace
- ✓ High-speed WiFi
- ✓ Complimentary coffee
- ✓ Locker access
- ✗ Private cabin
- ✗ Meeting room credits

Select Plan

Monthly Flex
Best for Freelancers
₹ **7,999** /mo

- ✓ Dedicated hot desk
- ✓ High-speed WiFi
- ✓ Unlimited coffee & tea
- ✓ Locker access
- ✓ 4 meeting room hrs/mo
- ✓ Community events

Select Plan

Private Office
For Teams 2-8
₹ **24,999** /r

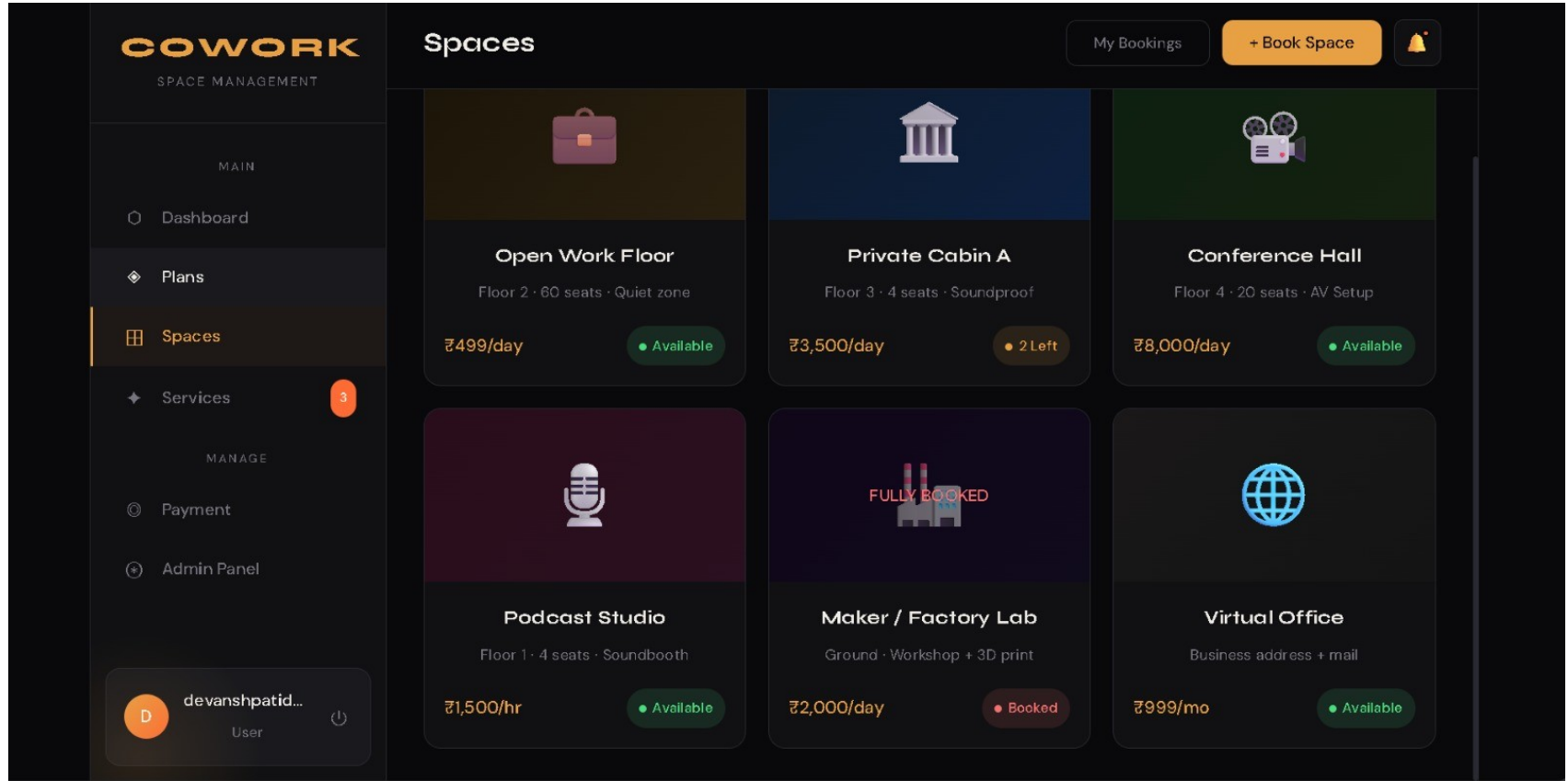
- ✓ Dedicated private cabin
- ✓ Business address
- ✓ 16 meeting room hrs/mo
- ✓ Catering add-ons
- ✓ Dedicated support staff
- ✓ GST invoice & reports

Select Plan

My Bookings + Book Space

devanshpatid... User

Screenshot - 4



The screenshot displays the COWORK Space Management interface. On the left is a dark sidebar with navigation options: Dashboard, Plans, Spaces (highlighted), and Services (with a red badge showing '3'). Below these are 'MANAGE' options: Payment and Admin Panel. At the bottom of the sidebar is a user profile for 'devanshpatid...' with a 'User' label and a power icon. The main content area is titled 'Spaces' and features a grid of six space cards. Each card includes an icon, a title, details, pricing, and availability status. Top right controls include 'My Bookings', '+ Book Space', and a notification bell.

Space Name	Details	Price	Status
Open Work Floor	Floor 2 · 60 seats · Quiet zone	₹499/day	Available
Private Cabin A	Floor 3 · 4 seats · Soundproof	₹3,500/day	2 Left
Conference Hall	Floor 4 · 20 seats · AV Setup	₹8,000/day	Available
Podcast Studio	Floor 1 · 4 seats · Soundbooth	₹1,500/hr	Available
Maker / Factory Lab	Ground · Workshop + 3D print	₹2,000/day	Booked
Virtual Office	Business address + mail	₹999/mo	Available

Screenshot - 5

COWORK
SPACE MANAGEMENT

MAIN

Dashboard

Plans

Spaces

Services 3

MANAGE

Payment

Admin Panel

devanshpatid...

User

Services & Add-ons

My Bookings

+ Book Space

**Food & Beverages**
Catering, snacks, meals
on demand
✓ Added

**Conveyance**
Cab booking, airport
transfers

**IT Support**
Tech setup, printer,
projector

**Secure Locker**
Daily or monthly locker
rental
✓ Added

**Event Hosting**
Host meetups,
workshops, launches

**Print & Scan**
Color / B&W printing,
scanning

**Mail Handling**
Receive and forward
courier

**Membership Perks**
Discounts, partner
benefits

Active Add-ons

SERVICE	PLAN	COST	STATUS
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Screenshot - 6

COWORK

SPACE MANAGEMENT

MAIN

Dashboard

Plans

Spaces

Services 3

MANAGE

Payment

Admin Panel

devanshpatid...

User

Make Payment

My Bookings

+ Book Space

Payment Method

Card

UPI

Net Banking

Wallet

CARDHOLDER NAME

Rahul Sharma

CARD NUMBER

4242 4242 4242 4242

EXPIRY

MM/YY

CVV

...

Pay ₹8,499 →

Order Summary

Monthly Flex Plan	₹7,999
Food Add-on	₹250
Secure Locker	₹500
Member Discount	- ₹250
Total	₹8,499

★

Gold Member

You'll earn 84 pts for this purchase

Future Scope

- AI Pricing Optimization: Predictive analytics for real-time rate discovery.
- Mobile Expansion: Native React Native apps for on-the-go booking.
- IoT Smart Access: Automated door locks and energy management.
- AR Workspace Tours: Immersive 3D virtual space previews.